Arjun Krishna Krishnakumar

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Irish Stamp 1G | *Open to work in Ireland*

PROFESSIONAL SUMMARY

Full Stack Developer and Automation Engineer with 3+ years of experience building production microservices, REST APIs, and data pipelines for global enterprise clients including Maersk and 150+ hospitals across India and the Middle East. Proficient in Python, TypeScript, React, PostgreSQL, and Docker, with strong backend engineering across ASP.NET Core and FastAPI. MSc in Data Science and Analytics (First Class Honours, MTU 2026) with applied machine learning, statistical modelling, and anomaly detection. Delivered a 10x query latency improvement on a live clinical scheduling system. Experienced building AI assisted applications including a RAG based document intelligence platform and a Model Context Protocol server for analytics tooling.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C#, SQL

Frontend: React, Next.js, Angular, Tailwind CSS, RxJS, Vite

Backend and APIs: FastAPI, ASP.NET Core (.NET 10), Node.js, Express.js, REST APIs, JWT, Microservices

Databases: PostgreSQL, SQL Server, MongoDB, MySQL

Data and Analytics: Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI, dbt, Apache Airflow, Apache Kafka

Cloud and DevOps: AWS, Docker, GitHub Actions CI/CD, Azure DevOps, Git, Linux

AI and ML: RAG (Retrieval Augmented Generation), Semantic Kernel, Applied Machine Learning, Statistical Modelling, Anomaly Detection, MCP (Model Context Protocol)

Practices: Microservices architecture, Agile/Scrum, code review, technical mentoring

WORK EXPERIENCE

Full Stack Developer | Cydez Technologies, Remote | January 2025 to July 2025

- Designed and built REST APIs for shipment creation, real time status tracking, and warehouse inventory within a microservices architecture serving Maersk and enterprise logistics clients; APIs processed thousands of daily transactions.
- Reduced data access latency across delivery, warehouse, and shipment modules by refactoring PostgreSQL and Entity Framework Core queries; applied clean code and separation of concerns principles throughout.
- Automated CI/CD pipelines using GitHub Actions and Docker; contributed to Azure DevOps configuration for a 20 person Agile/Scrum team.
- Reviewed code and mentored junior developers; participated in sprint planning and backlog refinement.

Software Engineer | Kameda Infologics, India | October 2022 to September 2024

- Developed backend microservices in ASP.NET Core and C# across patient management, clinical scheduling, and records modules serving 150+ hospitals in India and the Middle East under healthcare compliance requirements.
- Optimised PostgreSQL queries on the live patient scheduling module, reducing booking and rescheduling response times from over 10 seconds to under 1 second (10x improvement) on a high concurrency clinical platform.
- Built analytics dashboards for clinical and administrative stakeholders; gathered requirements through workshops and translated them into automated reports.
- Authored unit tests using Jasmine and Karma, and supported integration and end to end testing across multiple Agile teams.

Software Engineer | Luminar Technolab, India | November 2021 to October 2022

- Led 3 of 5 production releases on a 5 person team delivering clinical workflow applications; managed Linux server deployments including SSH, reverse proxies, and production diagnostics.
- Delivered full stack technical training to cohorts of 60 students covering Angular, React, TypeScript, Node.js, and REST APIs.

EDUCATION

MSc in Data Science and Analytics | Munster Technological University, Cork, Ireland | September 2025 to September 2026

First Class Honours (1.1) | *Data Mining*, *Applied Machine Learning*, *Statistical Modelling*, *Anomaly Detection*, *Scientific Programming in Python*

BTech in Mechanical Engineering | SCMS School of Engineering and Technology, India | January 2021
GPA: 8.05 / 10

PROJECTS

MediClaim AI | *React*, *TypeScript*, *Vite*, *Tailwind CSS*, *ASP.NET Core*, *PostgreSQL*, *FastAPI*, *Semantic Kernel*

- Built a full stack AI assisted insurance claims platform featuring PDF upload and text extraction with a custom chunking engine, RAG based question answering over policy documents via Semantic Kernel, rule based risk classification of claims (high, medium, low), JWT authenticated REST API, and a dashboard with document and claim analytics.
- Frontend built in React, TypeScript, and Vite with Tailwind CSS; backend in ASP.NET Core (.NET 10) with Entity Framework Core and PostgreSQL; document processing with PdfPig.

Pharmacy BI Dashboard | *Python*, *Pandas*, *Matplotlib*, *Seaborn*, *Tableau*

- End to end sales analytics pipeline on 50,000+ pharmacy transactions: data quality checks, revenue verification, exploratory analysis, and interactive Tableau dashboards covering branch performance, product profitability, customer demographics, and payment flow visualisation.

Analytics MCP Server | *Python*, *PostgreSQL*, *asyncpg*, *dbt*, *Apache Airflow*, *Apache Kafka*

- Built a Model Context Protocol (MCP) server exposing internal analytics data to AI coding tools (Cursor, Windsurf, Zed) via stdio. Implemented 5 MCP tools: query_metric, get_dashboard_data, generate_dashboard_code, run_dbt_model, and list_available_metrics; backed by PostgreSQL with raw and analytics schemas, dbt for incremental transformations, Airflow for nightly DAG scheduling, and Kafka for real time event ingestion.

CERTIFICATIONS

- AWS Certified Machine Learning Specialty (in progress, target December 2026)
- Supervised Machine Learning: Regression and Classification | Coursera / DeepLearning.AI